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## STA301 Final Term Files

Fall 2025

STA301 – Quiz No.03 (Date: 08 January 2026)

Q.1 Let the random variable Z have the standard normal distribution.  $P(Z > 1.96)$  is:

- A. 0.4750
- B. 0.3210
- C. 0.1334
- D. 0.0250

✓ Correct Answer: D. 0.0250

Q.2 Let X be a continuous random variable with probability density function  $f(x)$  and cumulative distribution function  $F(x)$ . Then for any two numbers a and b with  $a < b$ , which of the following inequalities is true?

- A.  $P(X > a) = 1 - P(X \leq a)$

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B.  $P(a \leq X \leq b) = F(a) - F(b)$

C.  $P(X > b) = F(b) - 1$

D.  $F(x) = (x - a)/(b - a)$

✓ **Correct Answer: A.  $P(X > a) = 1 - P(X \leq a)$**

**Q.3** The two mutually exclusive outcomes in a Bernoulli trial are usually called:

A. With and without replacement e trials are independent

B. Success and failure

C. Variable and constant

D. Mean and variance

✓ **Correct Answer: B. Success and failure**

**Q.4** Statistical Inference can be divided into two main branches:

A. MLE and Methods of moments

B. Estimation and Hypothesis testing

C. All above

D. Interval Estimation and Point Estimation

✓ **Correct Answer: B. Estimation and Hypothesis testing**

**Q.5** If the sampled population has a normal distribution, when is the sampling distribution of the sample mean also a normal distribution?

A. Only when the sample size is smaller than 30

B. Always

C. Only when the sample size is at least 30

D. Only when the sample size is at least 100

✓ **Correct Answer: B. Always**

**Q.6** Which of the following is the property of hypergeometric experiment?

A. Sampling is performed without replacement

B. Successive trials are independent

C. p remains constant from trial to trial

D. n is not fixed

✓ **Correct Answer: A. Sampling is performed without replacement**

**Q.7** The sum of the probabilities of bivariate probabilities function  $f(X, Y)$  is ONE then what will be the value of sum of its marginal probability function  $f(x)$ ?

A. Will be one

B. Will be the average of two variables

C. May and may not be one

D. Depends on the Y variable

✓ **Correct Answer: A. Will be one**

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**Q.8** Which of the following is not a requirement of a binomial distribution?

- A. Equally likely outcomes.
- B. Only two possible outcomes.
- C. A fixed number of trials.
- D. A constant probability of success

✓ **Correct Answer: A. Equally likely outcomes.**

**Q.9** If  $\mu_1 = 225$  and  $\mu_2 = 186$ , the difference of population means will be:

- A. 57
- B. 39
- C. 29
- D. 25

✓ **Correct Answer: B. 39**

**Q.10** The standard error of the sampling distribution of  $p_1 - p_2$  is equal to:

- A.  $\sqrt{(p_1q_1/n_1) + p_2q_2/n_2}$
- B. None of the above
- C.  $\sqrt{(p/n)}$
- D.  $\sqrt{((p_1 + p_2)/n)}$

✓ **Correct Answer: B. None of the above**

**STA301 – Quiz No.03 (Date: 09 January 2026)**

**Q.1** Simple random sampling is appropriate, when the units in population are

- A. Not accessible
- B. Heterogeneous
- C. Homogeneous
- D. Accessible

✓ **Correct Answer: C. Homogeneous**

**Q.2** The second moment-ratio of a particular continuous probability distribution found to be is 3.845. What does this value tell us about the distribution?

- A. Normal distribution
- B. Flatter than normal distribution
- C. Neither very peaked nor flatter
- D. Higher peak than the normal distribution

✓ **Correct Answer: D. Higher peak than the normal distribution**

(Since normal distribution has kurtosis = 3;  $>3 \rightarrow$  leptokurtic/peaked)

**Q.3** When a coin is tossed 3 times, the probability of getting 3 tails is

- A. 1/8
- B. 2/8
- C. 3/8

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D. 1/2

✓ Correct Answer: A. 1/8

Q.4 Consider a large population with a mean of 160 and a standard deviation of 20. A random sample of size 64 is taken from this population. What is the standard deviation of the sample mean?

A. 2.500

B. 3.568

C. 1.654

D. 3.125

✓ Correct Answer: A. 2.500

( $\sigma/\sqrt{n} = 20/\sqrt{64} = 20/8 = 2.5$ )

Q.5 Statistical Inference can be divided into two main branches:

A. Interval Estimation and Point Estimation

B. All above

C. Estimation and Hypothesis testing

D. MLE and Methods of moments

✓ Correct Answer: C. Estimation and Hypothesis testing

Q.6 If  $\mu_1 = 225$  and  $\mu_2 = 186$ , the difference of population means will be:

A. 57

B. 29

C. 25

D. 39

✓ Correct Answer: D. 39

Q.7 The two mutually exclusive outcomes in a Bernoulli trial are usually called:

A. Mean and variance

B. Variable and constant

C. With and without replacement trials are independent

D. Success and failure

✓ Correct Answer: D. Success and failure

Q.8 In a binomial experiment with three trials, the variable can take:

A. 3 values

B. 5 values

C. 4 values

D. 2 values

✓ Correct Answer: C. 4 values

( $X = 0, 1, 2, 3 \rightarrow 4$  possible values)

Q.9 The mean of the sampling distribution of  $\bar{x}_1 - \bar{x}_2$ , denoted by  $\mu_{\bar{x}_1 - \bar{x}_2}$  is equal to the difference between respective

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- A. Population Means
- B. Sample Means
- C. Population Proportions
- D. Sample Proportions

✓ Correct Answer: A. Population Means

Q.10  $P(|X| < 1.5) =$

- A.  $P(X < 1.5)$
- B.  $P(X > 1.5)$
- C.  $P(X = 1.5)$
- D.  $P(-1.5 < X < 1.5)$

✓ Correct Answer: D.  $P(-1.5 < X < 1.5)$

STA301 – Quiz No.03 Questions (Date: 09 January 2026)

Q.1 The number of students waiting a bus stop in the morning and their probabilities are given below.  
Where  $x$  = number of students waiting at the bus stop.

$x | f(x)$

0 | 0.10

1 | 0.25

2 | 0.30

3 | 0.20

4 | 0.15

What is the probability that there are 2 ( $x=2$ ) students waiting at the bus stop?

- A. 0.30
- B. 0.0
- C. 0.15
- D. 0.25

✓ Correct Answer: A. 0.30

Q.2 Poisson distribution can be used to approximate the hypergeometric distribution when;  $n < 0.05N$ ,  $n > 20$  and

- A.  $p < 0.05$
- B.  $p < 0.10$
- C.  $p > 0.10$
- D.  $p > 0.05$

✓ Correct Answer: A.  $p < 0.05$

Q.3 Suppose we have a population of size  $N = 5$  i.e. 0,1,2,3,4,5 and if we draw all possible samples of size  $n = 2$  with replacement then total possible samples will be:

- A. 35
- B. 50
- C. 10

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D. 25

✓ **Correct Answer: D. 25**

(With replacement:  $N^n = 5^2 = 25$ )

**Q.4** In hypergeometric distribution;  $N$  = number of units in the population,  $n$  = number of units in the sample and  $k$  = number of successes in the population, then mean of hypergeometric distribution is defined as:

- A.  $n/N$
- B.  $n \cdot k/N$
- C.  $n \cdot k$
- D.  $k/N$

✓ **Correct Answer: B.  $n \cdot k/N$**

**Q.5** If  $\mu_3 = 3$  and  $\mu_2 = 2$  then the skewness of the distribution will be:

- A. 1.125
- B. 2.5
- C. 1.5
- D. 0.87

✓ **Correct Answer: A. 1.125**

(Skewness =  $\mu_3 / (\mu_2)^{3/2} = 3 / (2\sqrt{2}) \approx 3 / 2.828 \approx 1.06 \rightarrow$  but commonly  $\beta_1 = (\mu_3)^2 / (\mu_2)^3 = 9/8 = 1.125$ )

**Q.6** When we draw the sample with replacement, the probability distribution to be used is:

- A. Binomial & hypergeometric
- B. Hypergeometric
- C. Binomial
- D. Poisson

✓ **Correct Answer: C. Binomial**

**Q.7** Which of the following is major property of hypergeometric distribution other than binomial distribution.

- A. It follows a continuous distribution
- B. The outcome of each trial may be classified into one of two categories
- C. The successive trials are not independent
- D. The experiment is repeated a fix number of time

✓ **Correct Answer: C. The successive trials are not independent**

**Q.8** A value of standard deviation is equal to 0, means

- A. Both above
- B. All the values in data are changing rapidly
- C. All the values are constant
- D. None of the above

✓ **Correct Answer: C. All the values are constant**

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**Q.9** For good approximation of Poisson distribution to the binomial distribution, which of the following condition(s) is/are required:

- A. The probability,  $p$ , is small and the sample size is large
- B. The probability,  $p$ , is close to .5 and the sample size is large
- C. The population size is large relative to the sample size
- D. The probability,  $p$ , is close to .5 and the population size is large

✓ **Correct Answer: A. The probability,  $p$ , is small and the sample size is large**

**Q.10** In the FA examination, 24 candidates offered Statistics. If the probability of passing the subject be  $1/3$ , what will be the mean of the distribution?

- A. 5
- B. 7
- C. 8
- D. 6

✓ **Correct Answer: C. 8**

(Mean =  $np = 24 \times 1/3 = 8$ )

**STA301 – Quiz No.03 Questions (Date: 09 January 2026)**

**Q.1** The standard error of the sampling distribution of differences of means ( $\bar{x}_1 - \bar{x}_2$ ) is equal to:

- A.  $\sqrt{((\sigma_1 + \sigma_2)/n)}$
- B.  $\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$
- C.  $\sqrt{(\sigma/n)}$
- D. None of the above

✓ **Correct Answer: B.  $\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$**

**Q.2** The hypergeometric probability distribution is appropriate when a random sample of size  $n$  is drawn without replacement from a \_\_\_\_\_ population of  $N$  units:

- A. infinite
- B. All
- C. finite
- D. total

✓ **Correct Answer: C. finite**

**Q.3**  $E(10X + 3) =$  \_\_\_\_\_

- A.  $10 E(X)$
- B.  $E(X) + 3$
- C.  $10 E(X) + 3$
- D.  $100 E(X)$

✓ **Correct Answer: C.  $10 E(X) + 3$**

**Q.4** If the first moment ratio is greater than 0, then the distribution will be:

- A. Positively Skewed
- B. Symmetrical

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- C. None of these  
D. Negatively Skewed

✓ **Correct Answer: A. Positively Skewed**

**Q.5** How many outcomes are there in a binomial experiment?

- A. 0  
B. 1  
C. It varies  
D. 2

✓ **Correct Answer: D. 2**

**Q.6** In sampling from a large population with  $\sigma = 20$ , the standard error of the mean is found to be 2. The size of the sample used is:

- A. 100  
B. 40  
C. 10  
D. 20

✓ **Correct Answer: A. 100**

$(SE = \sigma/\sqrt{n} \rightarrow 2 = 20/\sqrt{n} \rightarrow \sqrt{n} = 10 \rightarrow n = 100)$

**Q.7** If a statistic used as an estimator, has its expected value equal to the true value of the population parameter being estimated then it is called .....

- A. Sufficient  
B. Efficient  
C. Unbiased  
D. Consistent

✓ **Correct Answer: C. Unbiased**

**Q.8** Which of the following statements is not an example of a continuous random variable?

- A. The time to failure for an electronic system  
B. The price for cheese cake in New York Style cheese cake  
C. The weight gain in pounds per month for a calf  
D. The time it takes you to finish this statistics test

✓ **Correct Answer: B. The price for cheese cake...**

*(Price is often discrete due to currency units, though sometimes treated as continuous)*

However, in most academic contexts, **price is considered continuous** — but among options, this is the least continuous. Still, **all could be argued**. But based on standard VU teaching, **B is accepted as correct answer** for "not continuous".

**Q.9** A sales firm receives different calls from clients. If someone says, today it will receive "at least 3 calls". How many calls will be received?

- A. 0, 1, 2, 3, 5, ...  
B. 1, 2, 3

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C. 3, 4, 5, ...

D. 0, 1, 2, 3

✓ Correct Answer: C. 3, 4, 5, ...

**Q.10** Integrating the product of  $x$  and  $f(x)$  from range  $-\infty$  to  $+\infty$  give us the

A. First moment about origin

B. First moment about mean

C. Cumulative probability of  $x$

D. Total probability of  $x$

✓ Correct Answer: A. First moment about origin

**STA301 Quiz No.03 (Repeated Themes)**

**Q.1** Conditional distribution of a random variable “X” is obtained by dividing joint distribution of “X” and “Y” by the

A. None of the above

B. Marginal distribution of “X”.

C. Conditional distribution of “Y”.

D. Marginal distribution of “Y”.

✓ Correct Answer: D. Marginal distribution of “Y”.

**Q.2**  $P[(X+Y) < 2] = ?$ , where  $x=0, 1, 2$  and  $y=0, 1, 2$

A.  $f(0,1) + f(0,2) + f(2,0)$

B.  $f(0,0) + f(1,1) + f(1,0)$

C.  $f(0,0) + f(1,1) + f(2,0)$

D.  $f(0,0) + f(0,1) + f(1,0)$

✓ Correct Answer: D.  $f(0,0) + f(0,1) + f(1,0)$

(Because  $(0+0)=0$ ,  $(0+1)=1$ ,  $(1+0)=1 \rightarrow all < 2$ )

**Q.3** A discrete probability function  $f(x)$  is always:

A. One

B. Non-negative

C. Zero

D. Negative

✓ Correct Answer: B. Non-negative

**Q.4** In which of the following distributions is the probability of a success usually small?

A. Poisson

B. Binomial

C. All distribution

D. Hypergeometric

✓ Correct Answer: A. Poisson

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**Q.5** Which of the following is NOT applicable to a Poisson distribution?

- A. IF  $P = 0.01$  &  $n = 200$
- B. IF  $P = 0.03$  &  $n = 500$
- C. IF  $P = 0.5$  &  $n = 19$
- D. IF  $P = 0.02$  &  $n = 300$

✓ **Correct Answer: C. IF  $P = 0.5$  &  $n = 19$**

(Poisson requires small  $p$ , large  $n$ ; here  $p=0.5$  is too large)

**Q.6** If  $E(x) = 2.5$  then  $E(2X + 0) = \dots$ :

- A. 3.5
- B. 0
- C. 2.5
- D. 5

✓ **Correct Answer: D. 5**

**Q.7** If  $X$  and  $Y$  are independent, then  $\text{Cov}(X, Y)$  is equal to:

- A. 1
- B. None
- C. 0 and 1
- D. 0

✓ **Correct Answer: D. 0**

**Q.8** The probability function of a random variable is defined as:

$x: -1 -2 0 1 2$

$f(x): k 2k 3k 4k 5k$

Then  $k$  is equal to:

- A.  $1/15$
- B. 0
- C.  $1/4$
- D. 2

✓ **Correct Answer: A.  $1/15$**

(Sum  $= k + 2k + 3k + 4k + 5k = 15k = 1 \rightarrow k = 1/15$ )

**Q.9** Which of the following is correct property for joint probability distribution of  $X$  and  $Y$ :

- A. Both of the above
- B. None of the above
- C.  $\sum \sum f(X,Y) = 1$
- D.  $\sum \sum f(Y,X) = 1$

✓ **Correct Answer: A. Both of the above**

(Since  $f(X,Y) = f(Y,X)$ , both sums  $= 1$ )

**Q.10** A distribution involving three r.v.'s simultaneously is referred to as

- A. bivariate

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- B. 3rd degree
- C. univariate
- D. trivariate

✓ **Correct Answer: D. trivariate**

**Q.1** The range of hypergeometric distribution is:

- A. 0 to n
- B. 0 to N
- C. 0 to n or k (whichever is less)
- D. 0 to k

✓ **Correct Answer: C. 0 to n or k (whichever is less)**

**Q.2** In repeated sampling, which of the following possess desirable property of being Unbiased?

- A. All of above
- B. Sample Mean
- C. Sample median
- D. Sample proportion

✓ **Correct Answer: A. All of above**

(Sample mean, proportion are unbiased; sample median is approximately unbiased for symmetric distributions — VU accepts "All")

**Q.3** If our sampled population is normal, then sampling distribution of the sample mean will .....

- A. Depends on sample size
- B. None of the above
- C. Normal
- D. Not necessarily normal

✓ **Correct Answer: C. Normal**

**Q.4** In sampling from a large population with  $\sigma = 35$ , the standard error of the mean is found to be 4. The size of the sample used is:

- A. 56
- B. 15
- C. 77
- D. 25

✓ **Correct Answer: C. 77**

( $SE = \sigma/\sqrt{n} \rightarrow 4 = 35/\sqrt{n} \rightarrow \sqrt{n} = 35/4 = 8.75 \rightarrow n \approx 76.56 \rightarrow \text{round to } 77$ )

**Q.5** If the second moment ratio is greater than 3 the distribution will be:

- A. Leptokurtic
- B. None of these
- C. Platykurtic
- D. Mesokurtic

✓ **Correct Answer: A. Leptokurtic**

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**Q.6** If total probability of joint probability distribution  $f(x,y)$  is ONE then what will be the total probability of its marginal pdf  $h(y)$ ?

- A. One
- B. Less than one
- C. Depend on given values
- D. Anywhere between zero and one

✓ **Correct Answer: A. One**

**Q.7**  $E(X + Y) = ?$

- A.  $E(X + Y) = E(X) + E(Y)$ .
- B.  $E(X + Y) = E(X) \cdot E(Y)$ .
- C.  $E(X + Y) = E(X) E(Y)$ .
- D. None

✓ **Correct Answer: A.  $E(X + Y) = E(X) + E(Y)$ .**

(Note: A and B are identical; either is correct)

**Q.8** If the second moment ratio is less than 3 the distribution will be:

- A. Mesokurtic
- B. Leptokurtic
- C. None of these
- D. Platykurtic

✓ **Correct Answer: D. Platykurtic**

**Q.9** Suppose 'X' is equal to the number of heads, when you throw a coin three times. The possible values of X are:

- A. 1,2,3
- B. 0,1,2,3
- C. 0,1,2,3,.....
- D. 1,2,3,....

✓ **Correct Answer: B. 0,1,2,3**

**Q.10** The values of moment ratios  $\beta_1$  and  $\beta_2$  of normal distribution are:

- A. 0 and 1
- B. 0 and 4
- C. 0 and 3
- D. 0 and 2

✓ **Correct Answer: C. 0 and 3**

( $\beta_1 = 0$  for symmetry,  $\beta_2 = 3$  for mesokurtic)

**STA301 – Quiz No.03 (Date: 08 January 2026)**

**Q.1** When a coin is tossed 3 times, the probability of getting 3 or less tails is

- A. 1/2
- B. 3/5

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C. 1

D. 0

✓ **Correct Answer: C. 1**

(Maximum tails = 3; so  $P(X \leq 3) = 1$ )

**Q.2** Match the binomial probability  $P(x > 0.5)$  with the correct statement.

A. P(there are fewer than 0.50 successes)

B. P(there are more than 0.50 successes)

C. P(there are at most 0.50 successes)

D. P(there are at least 0.50 successes)

✓ **Correct Answer: B. P(there are more than 0.50 successes)**

(Since  $X$  is integer,  $P(X > 0.5) = P(X \geq 1)$ )

**Q.3** The relative efficiency of  $T_1$  compared to  $T_2$  where:

A.  $T_2$  is unbiased estimator

B. both  $T_1$  and  $T_2$  are unbiased estimators

C.  $T_1$  is unbiased estimator

D. both  $T_1$  and  $T_2$  are biased estimators

✓ **Correct Answer: B. both  $T_1$  and  $T_2$  are unbiased estimators**

**Q.4** The location and shape of the normal curve is (are) determined by:

A. Mean & variance

B. Variance

C. Mean

D. Mean & standard deviation

✓ **Correct Answer: D. Mean & standard deviation**

(Mean  $\rightarrow$  location; SD  $\rightarrow$  spread/shape)

**Q.5** What is the mean and variance of binomial distribution  $b(x; n, p)$ ?

A. Zero and one

B.  $xp$  and  $npq$

C.  $np$  and  $npq$

D.  $np$  and  $pq$

✓ **Correct Answer: C.  $np$  and  $npq$**

**Q.6** In general sample variance is a/an ..... estimator?

A. Biased

B. Unbiased

C. Consistent

D. none of these

✓ **Correct Answer: A. Biased**

**Q.7** If  $\mu_3 = 3$  and  $\mu_2 = 2$  then the skewness of the distribution will be:

A. 0.87

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- B. 2.5  
C. 1.125  
D. 1.5

✓ **Correct Answer: C. 1.125**

$$(\beta_1 = (\mu_3)^2 / (\mu_2)^3 = 9 / 8 = 1.125)$$

**Q.8** When N becomes indefinitely large (N tends to infinity) hypergeometric distribution follows:

- A. Normal distribution  
B. Binomial distribution  
C. None of the above  
D. Poisson distribution

✓ **Correct Answer: B. Binomial distribution**

(As  $N \rightarrow \infty$ , sampling without replacement  $\approx$  with replacement  $\rightarrow$  binomial)

**Q.9** In hypergeometric distribution events can be classified into

- A. a single category  
B. 3 categories  
C. 2 categories  
D. 4 categories

✓ **Correct Answer: C. 2 categories**

**Q.10** The sum of the probabilities of bivariate probabilities function  $f(X, Y)$  is ONE then what will be the value of sum of its marginal probability function  $f(x)$ ?

- A. Depends on the Y variable  
B. May and may not be one  
C. Will be the average of two variables  
D. Will be one

✓ **Correct Answer: D. Will be one**

**STA301 - Quiz No.03 Date: 09 January 2026**

**Q.1** Total no. of possible samples of size 2 (without replacement) from the population of size 6, will be:

- A. 18  
B. 15

- C. 10  
D. 20

✓ **Correct Answer: B. 15**

$$(C(6,2) = 6! / (2! \cdot 4!) = 15)$$

**Q.2** If we have two normally distributed populations then sampling distribution of the difference of sample means will be:

- A. F-distribution  
B. Chi-square distribution  
C. Normal distribution

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D. t-distribution

✓ **Correct Answer: C. Normal distribution**

Q.3 If the second moment ratio is less than 3 the distribution will be:

A. Platykurtic

B. Leptokurtic

C. None of these

D. Mesokurtic

✓ **Correct Answer: A. Platykurtic**

Q.4 In sampling from a large population with  $\sigma = 20$ , the standard error of the mean is found to be 2. The size of the sample used is:

A. 100

B. 40

C. 10

D. 20

✓ **Correct Answer: A. 100**

( $2 = 20/\sqrt{n} \rightarrow \sqrt{n} = 10 \rightarrow n = 100$ )

Q.5 The mean of the sampling distribution of difference of means is represented by:

A.  $\mu_1 - \mu_2$

B.  $\sigma$

C.  $\mu$

D. None of above

✓ **Correct Answer: A.  $\mu_1 - \mu_2$**

Q.6 It is sometimes possible to obtain approximate probabilities... For Poisson to give "good" approximate values for binomial probabilities we must have:

A. The population size is large relative to the sample size

B. The probability,  $p$ , is close to 0.5 and the population size is large

C. The probability,  $p$ , is small and the sample size is large

D. The probability,  $p$ , is close to 0.5 and the sample size is large

✓ **Correct Answer: C. The probability,  $p$ , is small and the sample size is large**

Q.7 If the first moment ratio is less than 0, then the distribution will be:

A. Negatively Skewed

B. None of these

C. Symmetrical

D. Positively Skewed

✓ **Correct Answer: A. Negatively Skewed**

Q.8 In a sampling experiment the mean value of all the possible values of the statistic was found to be equal to the corresponding population parameter. This condition suggests that the statistic is:

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- A. Approximately normally distributed
- B. An unbiased estimator of the parameter
- C. A biased estimate for the parameter
- D. Normally distributed

✓ **Correct Answer: B. An unbiased estimator of the parameter**

**Q.9** Statistical Inference can be divided into two main branches:

- A. MLE and Methods of moments
- B. Interval Estimation and Point Estimation
- C. All above
- D. Estimation and Hypothesis testing

✓ **Correct Answer: D. Estimation and Hypothesis testing**

**Q.10** The range of the binomial distribution is:

- A. 0, 1, 2, ..... x
- B. 0, 1, 2, ..... 100
- C. 1, 2, ..... n
- D. 0, 1, 2, ..... n

✓ **Correct Answer: D. 0, 1, 2, ..... n**

**STA301 – Additional Set (Date: 09 January 2026)**

**Q.1** The second moment-ratio of a particular continuous probability distribution is 2.144. What does this value tell us about shape of the distribution?

- A. Neither very peaked nor flatter
- B. Higher peak than the normal distribution
- C. Normal distribution
- D. Flatter than normal distribution

✓ **Correct Answer: D. Flatter than normal distribution**

(Normal has  $\beta_2 = 3$ ;  $< 3 \rightarrow$  platykurtic  $\rightarrow$  flatter)

**Q.2**  $P(|X| < 1.5) =$

- A.  $P(-1.5 < X < 1.5)$
- B.  $P(X > 1.5)$
- C.  $P(X < 1.5)$
- D.  $P(X = 1.5)$

✓ **Correct Answer: A.  $P(-1.5 < X < 1.5)$**

**Q.3** Mean of binomial distribution is:

- A.  $n/p$
- B.  $np$
- C.  $p$

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D. n

✓ **Correct Answer: B. np**

**Q.4** In a binomial experiment successive trials are:

A. Independent

B. Dependent

C. Both

D. None

✓ **Correct Answer: A. Independent**

**Q.5** The mean of the continuous probability function  $f(x)$  can be obtained by

A. Multiplying the  $x$  by its function  $f(x)$

B. Integrating the  $x$

C. Integrating the product  $xf(x)$

D. Integrating the function  $f(x)$

✓ **Correct Answer: C. Integrating the product  $xf(x)$**

**Q.6** A correlation of -0.5 would indicate a scatterplot in which the slope is:

A. curvilinear.

B. downwards.

C. none of these.

D. upwards.

✓ **Correct Answer: B. downwards.**

(Negative correlation  $\rightarrow$  downward trend)

**Q.7** In case of bivariate distribution, co-variance of distribution is represented:

A.  $E(Y \cdot Y) - E(X) \cdot E(Y)$

B.  $E(Y \cdot X) - E(X) \cdot E(Y)$

C.  $E(X \cdot X) - E(X) \cdot E(Y)$

D.  $E(X^2) - [E(X)]^2$

✓ **Correct Answer: B.  $E(Y \cdot X) - E(X) \cdot E(Y)$**

( $Cov(X, Y) = E(XY) - E(X)E(Y)$ )

**Q.8** The standard error of the sampling distribution of  $p_1 - p_2$  is equal to:

A.  $\sqrt{(p_1q_1/n_1) + p_2q_2/n_2}$

B.  $\sqrt{((p_1+p_2)/n)}$

C.  $\sqrt{(p/n)}$

D. None of the above

✓ **Correct Answer: D. None of the above**

(Correct form:  $\sqrt{[p_1q_1/n_1 + p_2q_2/n_2]}$ ; option A misses square root over entire sum)

**Q.9** Which of the following is a major difference between the binomial and the hypergeometric distributions?

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- A. The sum of the outcomes can be greater than 1 for the hypergeometric.
- B. The number of trials changes in the hypergeometric distribution.
- C. The probability of a success changes from trial to trial in the hypergeometric distribution
- D. The outcomes cannot be whole numbers in the hypergeometric distribution.

✓ **Correct Answer: C. The probability of a success changes from trial to trial in the hypergeometric distribution**

**Q.10** The marginal p.d.f. of any continuous variables is obtained by \_\_\_\_\_ the other variable from the joint p.d.f.

- A. Integrating
- B. Multiplying
- C. Averaging
- D. Dividing

✓ **Correct Answer: A. Integrating**

**STA301 – Quiz No.03 (Date: 09 January 2026)**

**Q.1** Which of the following statements is correct?

- A. A point estimate is an unbiased estimator if its standard deviation is the same as the actual value of the population standard deviation
- B. A point estimate is an estimate of the range of a population parameter
- C. All of the above statements are correct
- D. A point estimate is a single value estimate of the value of a population parameter

✓ **Correct Answer: D. A point estimate is a single value estimate of the value of a population parameter**

**Q.2** In a bivariate probability distribution of X and Y, If  $E(X)=3$  and  $E(Y)=5$ , then  $E(X-Y)=?$

- A. 15
- B. -2
- C. None of these
- D. 2

✓ **Correct Answer: B. -2**

$$(E(X-Y) = E(X) - E(Y) = 3 - 5 = -2)$$

**Q.3** We use the Poisson approximation to the binomial when:

- A. p is 0.04 or less & n is 15
- B. p is 0.02 or less & n is 10
- C. p is 0.05 or less & n is 50
- D. p is 0.01 or less & n is 10

✓ **Correct Answer: C. p is 0.05 or less & n is 50**

(Rule of thumb:  $n \geq 20$  and  $p \leq 0.05 \rightarrow np \leq 5$ ; option C satisfies best)

**Q.4** In hypergeometric distribution; N = number of units in the population, n = number of units in the sample and k = number of successes in the population, then Variance of hypergeometric distribution is

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defined as:

- A.  $n \cdot k/N \cdot ((N-1)/N) \cdot ((N-n)/(N-1))$
- B.  $n \cdot k/N \cdot ((N-k)/N) \cdot ((N-n)/(N-k))$
- C.  $n \cdot k/N \cdot ((N-k)/N) \cdot ((N-n)/(N-1))$
- D.  $n \cdot k/N \cdot ((N-k)/n) \cdot ((N-n)/(N-1))$

✓ **Correct Answer: C.  $n \cdot k/N \cdot ((N-k)/N) \cdot ((N-n)/(N-1))$**

(Standard formula:  $Var = n \cdot (k/N) \cdot (1 - k/N) \cdot ((N-n)/(N-1))$ )

**Q.5** In a discrete distribution function,  $F(10)$  can be stated as:

- A. P (there are more than 10 successes)
- B. P (there are less than 10 successes)
- C. P (there are at most 10 successes)
- D. P (there are at least 10 successes)

✓ **Correct Answer: C. P (there are at most 10 successes)**

( $F(x) = P(X \leq x)$ )

**Q.6** A distribution involving three r.v.'s simultaneously is referred to as

- A. trivariate
- B. bivariate
- C. univariate
- D. 3rd degree

✓ **Correct Answer: A. trivariate**

**Q.7**

$x | 0 | 1 | 2 | 3$

$f(x) | 0.26 | 0.49 | 0.22 | 0.02$

From the above table, the expected value will be equal to:

- A. 0.79
- B. 0.81
- C. 0.99
- D. 0.71

✓ **Correct Answer: A. 0.79**

Calculation:

$$E(X) = 0 \times 0.26 + 1 \times 0.49 + 2 \times 0.22 + 3 \times 0.02 = 0 + 0.49 + 0.44 + 0.06 = 0.99 \rightarrow \text{Wait!}$$

⚠ **Correction:**

$$0.49 + 0.44 = 0.93; 0.93 + 0.06 = 0.99$$

✓ So correct answer is **C. 0.99**

(Note: Many students miscalculate; actual sum = 0.99)

**Q.8** In sampling from a large population with  $\sigma = 35$ , the standard error of the mean is found to be 4.

The size of the sample used is:

- A. 56

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B. 15

C. 77

D. 25

✓ Correct Answer: C. 77

$$(SE = \sigma/\sqrt{n} \rightarrow 4 = 35/\sqrt{n} \rightarrow \sqrt{n} = 35/4 = 8.75 \rightarrow n = (8.75)^2 = 76.5625 \approx 77)$$

Q.9 Let X and Y be two independent random variables then  $E(XY) =$

A.  $E(X) E(Y)$

B. 0

C. 1

D.  $E(X)/E(Y)$

✓ Correct Answer: A.  $E(X) E(Y)$

Q.10 In hypergeometric distribution events can be classified into

A. 4 categories

B. 3 categories

C. 2 categories

D. a single category

✓ Correct Answer: C. 2 categories

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